

Hospital Appointment and Billing Analytics

Database Design

This project was developed to manage and analyze the day-to-day activities of a hospital. The main objective was to create a database that can efficiently store and organize information related to patients, doctors, appointments, treatments, bills, and payments.

A total of seven tables were created:

- Patients – stores patient information such as name, age, gender, city, and phone number.
- Departments – stores the different departments available in the hospital.
- Doctors – stores doctor details, specialization, and consultation fees.
- Appointments – stores appointment records between patients and doctors.
- Treatments – stores treatment details and treatment costs.
- Bills – stores billing information generated for patients.
- Payments – stores payment details and payment status.

Table Relationships

To ensure proper connectivity between the tables, foreign key relationships were implemented.

The relationships are:

- Doctors.department_id → Departments.department_id
- Appointments.patient_id → Patients.patient_id
- Appointments.doctor_id → Doctors.doctor_id
- Treatments.appointment_id → Appointments.appointment_id
- Bills.patient_id → Patients.patient_id
- Bills.appointment_id → Appointments.appointment_id
- Payments.bill_id → Bills.bill_id

Key Insights from Reports

- The hospital has patients from different cities.
- Cardiology, Neurology, Orthopedics, Pediatrics, and General Medicine departments are available.
- Some doctors handled more appointments than others.
- Both completed and cancelled appointments were recorded.
- The hospital generated revenue through treatments and consultation services.
- UPI, Card, Cash, and Net Banking were used as payment methods.
- Some bills were paid successfully, while a few remained unpaid.
- Patient billing reports helped track total bills and payments.
- The reports provided useful information for managing hospital operations more effectively.